

Joshua H. Hartlep

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EDUCATION

Duke University

Durham, NC

B.S. Electrical and Computer Engineering, B.S. Computer Science

May 2028

- GPA: **3.96/4.0**
- Relevant Coursework: Advanced Software Design, Data Structures & Algorithms, Computer Architecture, Machine Learning, Probability, Linear Systems of Differential Equations

PROFESSIONAL EXPERIENCE

Asana

May 2026 – Aug. 2026

Software Engineering Intern — AI Retrieval Team

New York, NY

- Building backend infrastructure for AI-powered product experiences, designing indexing and ranking components using efficient data structures over structured work graph entities
- Transforming complex application objects into searchable, LLM-readable representations for use in multi-tenant retrieval workflows
- Improving retrieval quality through hybrid lexical/vector search, combining keyword ranking with dense embeddings and evaluation-driven iteration

Bridgewater Associates

April 2026

1st Place Winner — AI Innovation Hackathon (\$10,000 Prize)

New York, NY

- Selected as 1 of 24 global participants for a sponsored AI immersion through a two-round interview process
- Built the **Paper Replication Machine**, a multi-agent application that parsed research PDFs, extracted methodology assumptions, and generated reproducible backtests
- Used LLM-assisted development workflows to rapidly prototype, debug, and review generated code, winning 1st Place for technical execution and product clarity

Worthington Enterprises

May 2025 – Aug. 2025

Software & Systems Engineering Intern

Chilton, WI

- Designed deterministic control and monitoring logic for automation, translating production constraints into technical requirements
- Built Python tooling to process real-time telemetry, detect anomalies, and support root-cause analysis for production downtime
- Applied statistical modeling to noisy data to improve reliability, yielding projected annual savings of **\$800K+**

PROJECTS

Shari POS (Full-Stack Ordering & Analytics System) | [GitHub](#) *FastAPI, Python, SQL, PostgreSQL, TypeScript*

- Built and deployed a full-stack web application supporting customer ordering, real-time order routing, and restaurant analytics workflows
- Designed relational database schemas and backend APIs, and frontend components for menu items, orders, sessions, kitchen routing, and customer-facing transaction flows
- Used Cursor and AI-assisted development tools to accelerate implementation while manually reviewing, testing, and refactoring generated code

DukeDine (AI-Powered Data Platform) | [Demo](#) *Python, AWS, PostgreSQL, pgvector, Node.js, Bedrock*

- Built a data platform aggregating structured performance metrics and unstructured dining data into searchable PostgreSQL-backed representations
- Designed ambiguity-handling logic for vision inputs, using clarify-or-commit flows to reduce misclassifications
- Optimized PostgreSQL ingestion pipeline with SHA-256 state tracking across 1,000+ entities, improving reliability of recurring data updates

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Python, C++, C, SQL, Verilog, Assembly

Web & Backend: React, Node.js, FastAPI, HTML/CSS, REST APIs, PostgreSQL, relational databases, pgvector

AI-Assisted Development: Cursor, Claude Code, LLM Prompting, Code Review, Agentic Workflows

Systems & Tools: Linux/Unix, Docker, Git, AWS, GitHub Actions, Telemetry, Metrics, Debugging